

Patent Infringement Detection Analysis — Indian Patent Application No. 201641043709 (Samsung Electronics)

TL;DR

- The single most likely literal infringement candidate identified is **Snapchat's "Dual Camera" feature (particularly the "Cutout" / green-screen layout, launched on iOS August 29, 2022 with Android rollout following in "the coming weeks/months" per Snap's creator blog and TechCrunch)**, which detects a user's face/body in the front-camera preview, extracts them, and composites the extract onto the live rear-camera preview — mapping closely onto Claims 1, 3, 4, and 6. All other candidates surveyed (BeReal, Instagram Dual, TikTok Now/Live Dual, Nokia Bothie, Huawei Dual-View, Xiaomi Dual Video, FiLMiC DoubleTake) implement split-screen or picture-in-picture composition without object detection/extraction across cameras and therefore likely do NOT literally infringe Claims 1-6.
- The entire AR virtual try-on category (ModiFace/L'Oréal, YouCam/Perfect Corp, Lenskart 3D Try-On, Warby Parker, Amazon AR, etc.) almost certainly does NOT infringe because these products use a SINGLE front camera plus pre-loaded 3D/2D product models — they do not detect or extract an object from a SECOND camera's live preview, which is the patent's signature limitation.
- Enforceability is severely constrained: prosecution status of IN 201641043709 could not be verified from official sources (the InPASS portal is CAPTCHA-gated and no third-party aggregator indexed this application number). Until grant is confirmed via InPASS, this analysis is hypothetical pre-screening only; the user must manually verify status on the Indian Patent Office portal before contemplating any enforcement action.

Key Findings

Phase 1 — Patent and Claim Breakdown

Patent overview. Application 201641043709 was filed at the Indian Patent Office on December 21, 2016, by Samsung Electronics Co., Ltd., with amended claims filed August 4, 2020 and prosecuted by Nitin Mohan Nair (Patent Agent 2585, identified externally as a senior partner/patent attorney at BananaIP Counsels, <https://www.bananaip.com/nitin-nair/>). The invention concerns generating a final camera preview by detecting an object in ONE camera's live preview (e.g., the rear), extracting it, and overlaying it on a DIFFERENT camera's live preview (e.g., the front), in real time.

Prosecution status: NOT VERIFIED. Despite targeted research, neither I nor a dedicated subagent could pull the live record from InPASS (<https://iprsearch.ipindia.gov.in/publicsearch> — CAPTCHA-gated and disallows automated fetching), and no third-party patent aggregator indexed this specific application number. The 2020 amended-claims filing is consistent with response to a First Examination Report (FER), placing the application in late-stage prosecution as of mid-2020, but no grant number, abandonment notice, or refusal could be located in open

sources as of May 21, 2026. **The user must manually query InPASS with the application number 201641043709 and the CAPTCHA before relying on any infringement conclusion.**

Family members. No PCT (WIPO) or US family member could be independently verified. The user's hypothesis that US20180103219A1 might be the US counterpart was not confirmable in this session and should be checked via USPTO Patent Center's foreign-priority field "IN 201641043709" or via Espacenet INPADOC. If no US/EP counterpart exists, this patent is enforceable only in India.

Claim element matrix (amended Aug 2020):

Element	Claim 1 (method)	Claim 2	Claim 3	Claim 4 (apparatus)	Claim 5	Claim 6	Claim 7 (system)
Detect object in camera preview (unit 202)	✓ literal	✓	✓	✓ literal	✓	✓	✓
Two cameras on same electronic device	✓	✓	✓	✓	✓	✓	✓
Select/extract object from cam 1's preview (unit 204)	✓ literal	✓	✓	✓	✓	✓	✓
Determine intended location in cam 2's preview (unit 206)	✓ literal	✓	— (stitching variant)	✓	✓	—	✓
Render by overlaying on intended location (unit 208)	✓ literal	✓ (property overlay)	—	✓	✓ (property overlay)	—	✓
Cross-camera asymmetry (source camera ≠ target camera)	✓ CRITICAL	✓	✓	✓ CRITICAL	✓	✓	✓
Live preview (not post-capture)	implied	implied	implied	implied	implied	implied	implied
Overlay PROPERTY (color/lighting) instead of whole object	—	✓	—	—	✓	—	✓

Element	Claim 1 (method)	Claim 2	Claim 3	Claim 4 (apparatus)	Claim 5	Claim 6	Claim 7 (system)
Stitch objects from both cameras	—	—	✓	—	—	✓	✓

The critical limitation, present in every claim, is the cross-camera asymmetry: the object must be selected from one camera's preview AND overlaid on the OTHER camera's preview. This single element disqualifies the entire AR virtual try-on industry (which uses one camera + a database) and all simple split-screen/ PiP dual-cam features (which do not detect or extract objects across cameras).

Phase 2 — Search Methodology Summary

I ran ~18 distinct web_search queries covering: (a) Indian Patent Office prosecution-status sources for 201641043709; (b) Samsung US/PCT family members; (c) every major dual-camera smartphone feature (Nokia Bothie/Dual Sight, Huawei P30 Dual-View, Xiaomi Dual Video, FiLMiC DoubleTake, Snapchat Dual Camera, TikTok Now/Live Dual, Instagram Dual/Candid Stories, BeReal); (d) AR virtual try-on architectures (ModiFace/L'Oréal, Lenskart 3D Try-On); (e) USPTO public PDF archive for related Samsung "preview overlay" patents. Sources fetched/cited include the Snapchat Lens Studio developer documentation (developers.snap.com), Snapchat Help Center, Elite Daily, Android Authority, MakeUseOf, Adweek, TechCrunch, PetaPixel, MacRumors, Macworld, CineD, Newsshooter, Filmic Pro official, Mi.com support, Huawei Central, Digital Trends, BGR, Recombu, L'Oréal official, Marketing Dive, BetaKit, ModiFace.com, WWD, Lenskart blog, IndiaAI government platform, USPTO public PDF archive, and an arXiv academic paper on multi-camera lighting estimation. A dedicated subagent verification of prosecution status returned UNKNOWN; the InPASS portal blocks automated lookup.

What was found: detailed technical implementation of every candidate dual-camera feature; confirmation that Snapchat's "Cutout" layout uses face/body segmentation from the selfie camera and composites it on the rear-camera feed. What was NOT found: official InPASS status; any litigation citing 201641043709; a confirmed PCT/US family member; any AR try-on product that uses two cameras with cross-camera extraction.

Phase 3 — Per-Candidate Infringement Assessment

Category A — AR Virtual Try-On Apps

A.1 ModiFace (L'Oréal subsidiary) — powering Sephora, Amazon, NYX, Maybelline, Lancôme, Garnier, A.S. Watson, etc.

- Description: AR makeup, hair, nail and skin try-on via web/app, deployed by L'Oréal brands and Amazon (US/Japan/Canada).
- Evidence: ModiFace.com product pages (<https://modiface.com/products-makeup.html>); L'Oréal corporate (<https://www.loreal.com/en/articles/science-and-technology/l-oreal-modiface-brings-ai-powered-virtual-makeup-try-ons-to-amazon/>); WWD

(<https://wwd.com/beauty-industry-news/color-cosmetics/amazon-implements-loreals-modiface-virtual-makeup-try-ons-1203148094/>).

- Technical implementation: "Amazon customers in the U.S. and Japan can now use the front-facing camera on their mobile phones to digitally try on lipstick shades in a live video of themselves or on a selfie" (L'Oréal/WWD verbatim). The makeup shade is rendered from a product database, not extracted from another camera feed.
- Claim analysis: Claim 1: NO — only one camera (front) is used. Claim 2 (property overlay — color): NO under claim 1 dependency, because Claim 2 still requires the property to come from a different camera's preview per Claim 1 base. Claim 3 (stitch from both cameras): NO. Claims 4-7: NO.
- Literal: NONE. Doctrine of equivalents: NONE — substituting a database-stored shade for an extracted preview object is not insubstantially different; the patent specifically distinguishes itself by sourcing the property from a live camera preview.
- Severity: **NONE**. Confidence: **High**.

A.2 Perfect Corp — YouCam Makeup / YouCam Nails / YouCam Hair

- Description: Taiwan-headquartered AR beauty platform used by Estée Lauder, MAC, Target, Maybelline, etc.
- Implementation parallels ModiFace: single front-camera tracking with pre-loaded SKU rendering (corroborated by Banuba comparison review at banuba.com/blog/best-virtual-makeup-try-on-solutions).
- Claim analysis: same as ModiFace.
- Severity: **NONE**. Confidence: **High**.

A.3 Lenskart 3D Try-On (India) — high priority jurisdiction

- Description: Lenskart is India's largest eyewear retailer by store count and revenue, per Business Standard (Oct. 2025): "It now operates India's largest eyewear retail network ... Lenskart reported revenue from operations of ₹6,652.5 crore [FY25], up 22.5% from ₹5,427.7 crore in FY24." Per Kotak Neo analysis citing India Retailing, "As of late 2024, Lenskart serves around 5 million customers and operates more than 2,500 stores worldwide, with approximately 2,000 of those outlets located in India." (Business Standard) (Kotak Neo)
- Evidence: Lenskart blog (<https://spectacular-blog.lenskart.com/spectacular-blog-lenskart-com-basic-guide-to-use-lenskart-virtual-ar-tool/>); IndiaAI government platform (<https://indiaai.gov.in/article/lenskart-reimagined-the-way-we-look-at-eyeware-thanks-to-its-ai-chops>); Twimbit; markhub24.com case study.
- Technical implementation per Lenskart blog: "Clicking the banner will turn on your front camera and you will notice a face scanner." User is scanned; AI face analysis recommends frames from Lenskart's catalogue; 3D models of frames are overlaid via the front camera only. (Lenskart)

- Claim analysis: Claim 1: NO — uses only the front camera; 3D frame models come from Lenskart's product database, not the rear camera's preview. The patent's cross-camera asymmetry is not met. Claims 2-7: NO.
- Severity: **NONE**. Confidence: **High**. (Despite Lenskart being the most India-relevant AR target, the technical architecture does not match the claims.)

A.4 Warby Parker / Zenni / GlassesUSA / FittingBox / Amazon AR shoes & lipstick / Pinterest Try-On / Google Shopping AR / Nykaa / Tira / Myntra / Tata CLiQ

- All independently sourced product pages and trade press indicate single-camera AR with pre-loaded product 3D models or shader-based effects. No source surfaced any retail-AR feature that captures a real-world product through the rear camera and overlays it on the user via the front camera in live preview.
- Severity: **NONE** for the entire retail-AR try-on category. Confidence: **High** (architectural pattern is industry-wide).

Caveat for the category: I did NOT find evidence of any production retail AR try-on that uses two simultaneous camera feeds with cross-camera object extraction. The closest academic concept is the arXiv paper "Multi-Camera Lighting Estimation for Photorealistic Front-Facing Mobile Augmented Reality" (<https://arxiv.org/pdf/2301.06143>) — but that paper uses the rear camera only for ambient lighting estimation, not for object extraction, and it is a research prototype, not a deployed product. (arxiv)

Category B — Smartphone Dual-Camera Simultaneous Modes (excluding Samsung)

B.1 Nokia "Bothie" / Dual Sight (HMD Global, Nokia 8 series, launched 2017)

- Description: Simultaneous front + rear camera capture in split-screen or picture-in-picture.
- Evidence: Digital Trends (<https://www.digitaltrends.com/phones/how-to-take-a-bothie-nokia/>); BGR India (<https://www.bgr.in/news/nokia-8-heres-how-the-bothie-dual-camera-setup-works-501403/>); Recombu review.
- Technical analysis: Pure split-screen / PiP composition. No object detection in either feed; no extraction; no cross-camera overlay. Per Digital Trends: "Dual Sight ... uses the Nokia 8's front and rear 13MP camera lenses simultaneously to present a split-screen image" — there is no detection or extraction step. (Carphonewarehouse)
- Claim analysis: Claim 1: NO (no object detection unit, no extraction, no overlay on a different camera's preview). Claim 3/6 (stitching): PARTIAL only in the broadest sense — the two feeds are composited side-by-side, which arguably is "combined" but does not "stitch at least one object" from each preview; the patent specification's "stitching" example envisions stitching a selfie INTO the rear camera's panoramic shot as a unified scene, not side-by-side juxtaposition.
- Literal infringement: NONE. Doctrine of equivalents: LOW probability.
- Severity: **LOW** (only Claim 3/6 with weak DoE argument). Confidence: **High** for non-infringement of Claims 1, 2, 4, 5, 7.

- Jurisdiction: HMD Global sells in India; Nokia 8 was launched 2017 pre-amendment but is sold globally.

B.2 Huawei Dual-View Video (P30/P30 Pro, EMUI 9.1, 2019)

- Evidence: Huawei Central (<https://www.huaweicentral.com/how-to-use-dual-view-video-mode-on-huawei-p30-and-p30-pro/>); SoyaCincau (<https://soyacincau.com/2019/03/22/huawei-p30-dual-view-video-recording/>).
- Technical: Uses two REAR cameras (telephoto + ultra-wide) simultaneously, NOT front + rear. Split-screen only. SoyaCincau verbatim: "Instead of using front and back cameras, the P30 is a split-screen video recording that uses the rear-mounted telephoto and ultra-wide-angle cameras." (SoyaCincau)
- Claim analysis: Claim 1 limitation "first camera and second camera" — arguably satisfied (two cameras on same device, both rear), but NO object detection/extraction/overlay step.
- Severity: **NONE**. Confidence: **High**.

B.3 Xiaomi/Redmi Dual Video Mode (Mi 10T Pro, Mi 11 Ultra, 12, 14T, 14T Pro, etc.)

- Evidence: Xiaomi global support (<https://www.mi.com/global/support/article/KA-36382/>).
- Implementation: Split-screen or PiP front + rear. Mi.com verbatim: "Dual video mode splits the scene on the viewfinder into two views. You can view and record a scene with the rear and front camera simultaneously." (MI)
- Claim analysis: No object detection/extraction. Pure composition.
- Severity: **NONE** for Claims 1, 2, 4, 5, 7; **LOW** for Claim 3/6 under weak DoE. Confidence: **High**.

B.4 LG / Oppo / OnePlus / Vivo / Honor / Realme / Motorola dual-camera simultaneous modes

- Based on industry pattern (no source contradicts), all implement composition without cross-camera object extraction.
- Severity: **NONE** / **LOW** (Claim 3/6 DoE only). Confidence: **Medium** (specific implementations for each OEM were not separately verified in this session).

B.5 Apple iPhone Multi-Cam (iOS 13+) and FiLMiC DoubleTake (iOS, Jan 2020)

- Evidence: Filmic Pro official (<https://www.filmicpro.com/products/doubletake/>); MacRumors (<https://www.macrumors.com/2020/01/28/doubletake-multi-camera-ios-app/>); Macworld; CineD; Newsshooter.
- Implementation: Records or previews any two iPhone lenses simultaneously in Picture-in-Picture, Split View, or Discrete (separate files) modes. Per Macworld: split-screen or PiP video, no post-production necessary. No object detection or extraction across cameras.
- Apple's native iOS Camera app does not (as of public information) implement cross-camera object extraction either; multicam recording via AVMultiCamPip is composition-only at the framework level.

- Claim analysis: Claim 1: NO. Claim 3/6 (stitching): LOW DoE.
- Severity: **NONE / LOW**. Confidence: **High**.
- Jurisdiction: Apple and Filmic are non-Indian; iPhones are sold in India.

Category C — Live AR Camera Filters

C.1 Snapchat Dual Camera — particularly "Cutout" layout (launched on iOS Aug. 29, 2022; Android rollout was slated for "the coming weeks" per Snap's creator blog and "the coming months" per TechCrunch on Aug. 29, 2022)

THIS IS THE HIGHEST-SEVERITY CANDIDATE IDENTIFIED.

- Description: Snapchat's Dual Camera lets users capture content using both the front and rear cameras simultaneously in four layouts: Vertical, Horizontal, Picture-in-Picture, and **Cutout**. (Elite Daily)
- Evidence (multiple independent sources):
 - Snapchat Help Center (<https://help.snapchat.com/hc/en-us/articles/8132982011668>): "Dual Camera allows you to capture content using the front and back cameras at the same time." (Snapchat Support)
 - Elite Daily (<https://www.elitedaily.com/news/snapchat-dual-camera-how-to-use>, Aug 2022): "Cutout works by detecting your face and body in the selfie camera and removing your background. So, you become the cutout on top of the front-facing camera." (Elite Daily)
 - MakeUseOf (<https://www.makeuseof.com/snapchat-bereal-dual-camera-feature/>): "Cutout creates a green-screen effect — superimposing the image from your selfie camera over the main background image from your rear camera." (MakeUseOf)
 - ScreenRant (<https://screenrant.com/snapchat-dual-camera-mode-use-how-explained/>): "the software edits out the background of the selfie photo/video and superimposes the user's face on the feed from the main camera." (ScreenRant)
 - Android Authority (<https://www.androidauthority.com/snapchat-dual-camera-3202616/>): "The cutout option inserts your body on top of the rear-facing video." (Android Authority)
- Snap Lens Studio Dual Camera developer documentation (<https://developers.snap.com/lens-studio/features/camera/dual-camera>): "The Dual Camera feature provides the capability to render both front and rear cameras at the same time ... extend powerful features like face and body tracking to custom textures, including the reverseCameraTexture ... creating different effects for front and back cameras simultaneously or tracking objects in pre-recorded media." (Snap)
- Claim-by-claim:
 - **Claim 1 (method):** YES, literal mapping appears strong for the Cutout layout: (i) detect object (user's face/body) in camera preview ✓ — Snapchat segments face/body from front-camera preview; (ii) two cameras on same device ✓ ; (iii) select/extract object from camera 1's preview ✓ — body cutout extracted from selfie camera; (iv) determine intended location in camera 2's preview ✓ — overlay is placed

on the rear-camera live preview (typically full-screen); (v) render final preview by overlaying ✓ — live composite is rendered before record/capture. **YES, likely literal infringement.**

- **Claim 2 (property overlay):** Cutout overlays the entire body cutout (an object), not a property. NO under Claim 2.
- **Claim 3 (stitching):** YES, arguably — the cutout from the front camera is stitched into the rear camera's preview scene to form a single integrated final preview, akin to the patent's described "selfie stitched into panoramic shot" example.
- **Claim 4 (apparatus):** YES — Snapchat-enabled smartphones running Snapchat constitute the "electronic device" with the recited units (object detection, extraction, location detection, rendering); the apparatus claim covers devices configured to perform Claim 1's steps.
- **Claim 5:** NO (property overlay not implemented).
- **Claim 6 (apparatus + stitching):** YES, parallel to Claim 3.
- **Claim 7 (system):** YES if reading on the Snapchat client + device combination.
- **Literal infringement analysis:** For the **Cutout** layout specifically, Snapchat appears to satisfy every element of Claim 1 literally. The other three layouts (Vertical, Horizontal, PiP) are mere composition and do NOT detect/extract; only Cutout uses segmentation. Snapchat's Lens Studio Dual Camera also makes the reverse camera texture available to Lens creators, enabling third-party Lenses that detect objects in one camera and apply effects/overlays referenced to the other — broadening the potential infringement set.
- **Doctrine of equivalents:** Not necessary; literal reading appears available.
- **Severity:** **HIGH** for Cutout layout and any third-party Snapchat Lens that uses cross-camera object placement; **LOW/NONE** for the other Snapchat layouts.
- **Confidence:** **Medium-High** based on multiple consistent secondary sources. To raise to "Very High," direct reverse engineering or Snap's technical documentation of the exact segmentation pipeline would be needed; Snap's own developer documentation explicitly describes "tracking objects" with the reverseCameraTexture, substantiating the technical implementation.
- **Jurisdiction:** Snap Inc.'s official newsroom (newsroom.snap.com/india-mau-250m) reports: "Snapchat announced its community in India has surged to over 250 million monthly active users (MAU), marking a significant milestone." Indian users can use Dual Camera Cutout, making this directly enforceable in India if the patent is granted. (snap)
- **Caveats:** (a) iOS rollout was Aug 29, 2022; Android followed within weeks/months; Cutout availability may vary by device. (b) The patent's "object" example is a spectacles/lipstick (small object), but Claim 1 says "at least one object" without size limitation; a body/face cutout qualifies.

C.2 TikTok Live Dual Camera (and TikTok "Now," announced September 16, 2022 per Visualistan: "TikTok has announced a new feature that enables and prompts users to take and share a photo or video captured by their device's front and back camera simultaneously ... called 'TikTok Now'")

- Evidence: TikTok discovery pages (<https://www.tiktok.com/discover/how-to-have-front-and-back-camera-on-tiktok-live>).
- Implementation: Allows livestream hosts to show front and back camera simultaneously, one window big and one small (PiP), with the ability to swap which camera is primary. TikTok
- Claim analysis: Pure PiP composition; no evidence of object detection or extraction across cameras. (No source surfaced a "cutout" equivalent for TikTok Live.)
- Severity: **NONE**. Confidence: **Medium** (TikTok's documentation is less specific than Snapchat's; haven't confirmed absence of segmentation feature exhaustively).

C.3 Instagram "Dual" feature (Reels) and "Candid Stories"

- Instagram Dual was announced July 21, 2022 in a tweet by Instagram CEO Adam Mosseri (@mosseri) listing 'Dual Camera' among new Reels features, and reported by PetaPixel on July 29, 2022 ("Instagram Launches New 'Dual' Camera Feature, Similar to BeReal"). Candid Stories launched December 2022. PetaPixel
- Evidence: PetaPixel (<https://petapixel.com/2022/07/29/instagram-launches-new-dual-camera-feature-similar-to-bereal/>); TechCrunch (<https://techcrunch.com/2022/07/28/instagram-dual-camera-feature-copies-bereal-misses-point/>); MakeUseOf.
- Implementation: Picture-in-picture front + rear; user can move/resize the small front-camera frame. No segmentation/cutout reported. VICE
- Severity: **NONE**. Confidence: **High**.

C.4 BeReal

- Implementation: PiP front + rear capture at notification time.
- Severity: **NONE**. Confidence: **High**.

C.5 Other AR filter platforms (Spark AR / Meta camera effects, Google Lens AR, Apple Memoji/Animoji, CapCut)

- All use a single camera (usually front) plus computer-generated overlays from the application — not extraction from a different camera.
- Severity: **NONE**. Confidence: **High** for the general architecture.

Category D — E-commerce / Retail AR (Indian focus)

Reviewed Lenskart (covered above); general findings indicate Nykaa, Tira, Myntra, Tata CLiQ, Flipkart use single-camera try-on with database products. Tira Beauty and Nykaa specifically white-label or partner with ModiFace/Perfect Corp class providers. None implement cross-camera object extraction. **Severity: NONE. Confidence: Medium-High** (individual Indian retailer specs were not separately verified at depth).

Phase 4 — Final Summary Table

Company / Product	Category	Claim(s) potentially infringed	Severity	Confidence	Jurisdiction relevance	Key evidence
Snapchat Dual Camera (Cutout layout)	Live AR filters	1, 3, 4, 6, 7	HIGH	Medium-High	Global incl. India (250M+ MAU)	Snap Help, Elite Daily, MakeUseOf, Android Authority, Snap Lens Studio docs
Snap Lens Studio dual-camera Lenses (3rd party)	Dev platform	1, 4, 7 (potentially 3/6)	MEDIUM	Medium	Global	developers.snap.com Dual Camera page
Nokia "Bothie" / Dual Sight	Smartphone OEM	3, 6 (weak DoE only)	LOW	High	Global incl. India	Digital Trends, BGR
Huawei Dual-View Video (P30)	Smartphone OEM	none	NONE	High	Limited in India	Huawei Central, SoyaCincau
Xiaomi Dual Video Mode	Smartphone OEM	3, 6 (weak DoE)	LOW	High	India	Mi.com support
Apple iOS multi-cam / Filmic DoubleTake	Smartphone / app	3, 6 (weak DoE)	LOW	High	Global incl. India	Filmic, MacRumors, Macworld
Instagram Dual / Candid Stories	Social app	none	NONE	High	Global	PetaPixel, TechCrunch
TikTok Now / Live Dual Camera	Social app	none	NONE	Medium	Global	TikTok, Visualistan

Company / Product	Category	Claim(s) potentially infringed	Severity	Confidence	Jurisdiction relevance	Key evidence
BeReal	Social app	none	NONE	High	Global	TechCrunch, PetaPixel
ModiFace / L'Oréal (Amazon, Sephora, etc.)	AR try-on	none	NONE	High	Global	L'Oréal, WWD, ModiFace
Perfect Corp / YouCam	AR try-on	none	NONE	High	Global	Banuba review
Lenskart 3D Try-On	Indian e-comm	none	NONE	High	India (priority)	Lenskart blog, IndiaAI, Business Standard
Warby Parker / Zenni / GlassesUSA / FittingBox	AR try-on	none	NONE	Medium-High	Global	industry pattern
Nykaa / Tira / Myntra / Flipkart AR	Indian retail	none	NONE	Medium	India	industry pattern

Phase 5 — Key Findings and Caveats

Most likely infringer. Snapchat's Dual Camera "Cutout" layout (and the broader Snap Lens Studio Dual Camera developer feature which enables third-party Lenses to extract objects from one camera and apply them to the other) is the single product where every claim element of independent Claim 1 appears literally satisfied. This finding should be the primary focus of any follow-on technical due diligence and legal opinion. Outside Cutout, no other consumer product surveyed shows the patent's signature cross-camera object detection-extraction-overlay architecture.

Why the bulk of the surveyed market does NOT infringe. Two architectural patterns dominate the market and both fall outside the claims:

1. *AR virtual try-on (ModiFace, Perfect Corp, Lenskart, Warby Parker, Amazon, Pinterest, Google, etc.) uses ONE camera plus a product database.* The patent specifically requires the

overlay source to be a SECOND camera's preview, not a stored asset.

2. *Smartphone dual-camera modes (Nokia Bothie, Huawei Dual-View, Xiaomi Dual Video, LG/Oppo/OnePlus dual modes, FiLMiC DoubleTake, BeReal, Instagram Dual, TikTok Live Dual, the non-Cutout Snapchat layouts) implement composition (split-screen or PiP) without object detection or extraction.* The patent's claims plainly require an object-level detection-extract-overlay pipeline that goes beyond mere frame composition.

Patent validity concerns. Several pre-priority-date publications discuss multi-camera preview architectures (e.g., US patent 9,307,153 "Method and apparatus for previewing a dual-shot image," which describes overlaying a partial image from a front sensor onto a rear image based on subject recognition — this is conceptually close to the Samsung Indian claim and predates it). A serious validity attack on novelty/obviousness could be mounted; the user should consider a thorough prior-art search if planning enforcement. [uspto](#)

Enforceability limitations.

- *Status risk.* The prosecution status of IN 201641043709 could not be verified. If the application is not granted, no claim can be asserted. Manual InPASS lookup is required.
- *Jurisdiction.* The patent is Indian. No US/PCT family was confirmed. Enforcement against US-only operations (e.g., Snap Inc. in the US, Filmic, ModiFace) requires equivalent US patent rights, which were not confirmed in this study.
- *Live preview vs. captured content.* The claim language repeatedly uses "preview" and "rendering" — strongly implying live, pre-capture rendering. Post-capture editing likely falls outside the claims even if it performs the same operations.

Details

Critical implementation differences that decide infringement

1. **Source of the overlaid object.** The patent requires the object to come from a CAMERA PREVIEW. Pre-loaded 3D/2D models (lipstick shades, eyeglass frames, makeup, jewellery, hairstyles) are NOT camera-preview-sourced and therefore fall outside Claims 1, 4, and dependents. This excludes the entire retail AR try-on category.
2. **Cross-camera asymmetry.** The patent requires the object's source camera preview $\neq$ the target camera preview. This excludes split-screen and PiP dual-cam features that merely show both feeds without object extraction.
3. **Object-level extraction.** The patent's object-extraction-unit step is more than alpha-blending — it implies detecting and isolating a specific object. Only Snapchat Cutout (face/body segmentation) and similar segmentation-based modes meet this.
4. **Live preview rendering.** All cited candidate features are live-preview, so this limitation is met for any candidate that satisfies the other three.

Why Snapchat Cutout maps so cleanly

When a user activates Cutout, the device's selfie camera runs in real time (camera preview #1).

Snap's segmentation model (object detection unit) identifies the user's face/body (the at-least-one object). The body silhouette is extracted (object extraction unit). The rear camera preview (camera preview #2) is rendered as the background; Snapchat determines where in that scene to position the cutout (location detection unit) — typically centered or at the lower part of the frame to mimic a person standing in the scene. The final composite preview, with the cutout overlaid live on the rear camera scene, is rendered to the user before they tap shutter (rendering unit). This sequence is essentially the literal recitation of Claim 1.

What was NOT analyzed (acknowledged gaps)

- Samsung's own Galaxy products were excluded per task instructions (Samsung is the patentee).
- Individual Chinese OEM dual-cam modes (Vivo, Oppo, Honor, Realme) were not separately verified at the product-page level beyond the general industry pattern.
- Snapchat Spectacles' AR overlay architecture (which involves a head-worn camera + the phone) was not analyzed in depth.
- I could not verify the prosecution status of 201641043709 from InPASS due to portal restrictions.
- I could not independently confirm or refute that US20180103219A1 is the US family equivalent.

Recommendations

Stage 1 — Status verification (do this first). Before any further action, manually query InPASS (<https://iprsearch.ipindia.gov.in/publicsearch>) with application number 201641043709 and the CAPTCHA. Confirm (a) GRANTED vs. PENDING vs. ABANDONED/REFUSED; (b) grant number if granted; (c) renewal-fee status. Concurrently search USPTO Patent Center for foreign-priority "IN 201641043709" and check Espacenet INPADOC for the full family. **Trigger to advance to Stage 2:** confirmation of grant in force in India OR confirmation of a granted US/EP family member with corresponding claims.

Stage 2 — Snapchat-focused technical deep dive. If grant is confirmed, commission a reverse-engineering or technical-investigation report on Snapchat's Dual Camera Cutout layout to confirm: (i) that face/body segmentation runs on the SELFIE camera preview (not the rear); (ii) that the cutout is composited onto the LIVE rear camera preview (not a freeze frame); (iii) that this happens BEFORE the user captures/records. Also evaluate third-party Snap Lenses that use Lens Studio's Dual Camera + reverseCameraTexture features. **Trigger to advance to Stage 3:** confirmed segmentation-based cross-camera overlay in live preview.

Stage 3 — Validity diligence. Concurrently with Stage 2, conduct a thorough prior-art search focused on (a) US 9,307,153 (LG-type "previewing dual-shot image"); (b) any pre-Dec-2016 publications by Microsoft, Snap, Facebook, or academic computer-vision groups describing cross-camera live segmentation and overlay; (c) the long-tail of camera-app patents from 2014–2016. **Trigger to advance to Stage 4:** validity reasonably defensible.

Stage 4 — Engagement. If validity holds, consider a formal opinion of counsel and, if commercially warranted, an Indian-court infringement action or licensing approach to Snap Inc. With more than 250 million monthly active users in India per Snap's own newsroom, the Indian market is directly relevant. Other lower-severity targets (Nokia/HMD Global, Xiaomi, OEMs with Claim 3/6 DoE exposure) are NOT recommended for enforcement priority because the DoE arguments are weak.

What would change these recommendations:

- If InPASS shows the application is abandoned or refused → halt analysis; the patent is not enforceable.
- If a strong, granted US family member is identified → expand enforcement scope to US Snap operations, Filmic Inc., and ModiFace's US deployments (though only Snap remains a meaningful target there).
- If reverse engineering reveals Snapchat Cutout actually uses a static photo or runs only on saved media (not live preview) → severity drops from HIGH to LOW.
- If a strong pre-2016 prior-art reference invalidates Claim 1 → patent unenforceable.

Caveats

- **This is a technical pre-screening, not legal advice or a formal infringement opinion.** Infringement is ultimately a legal determination requiring claim construction under Indian patent law (or the law of any other jurisdiction in which a family member is granted), discovery, expert testimony, and judicial interpretation.
- **Prosecution status of IN 201641043709 is unverified.** All infringement analysis is hypothetical until grant is confirmed.
- **Implementation details of Snapchat Cutout are inferred from public secondary sources** (Snap Help Center, Snap Lens Studio developer documentation, and consistent independent tech-press descriptions). The exact algorithmic pipeline is proprietary; the High severity rating could be revised after technical verification.
- **Minimum two-source corroboration was met** for every HIGH/MEDIUM finding (Snapchat: Snap Help Center, Elite Daily, MakeUseOf, ScreenRant, Android Authority, Snap Lens Studio docs — all independent). For NONE findings in retail AR, the industry-wide architectural pattern is corroborated by L'Oréal, WWD, Marketing Dive, BetaKit, ModiFace.com, Banuba comparison, Lenskart blog, IndiaAI.
- **Future-tense / hedged claims in sources.** Some marketing copy uses superlatives ("first ever," "industry first") that should not be relied on as fact; I have flagged the architectural pattern, not the marketing claims.
- **No litigation citing 201641043709** has been located in open sources; Samsung has not (as far as can be determined) publicly asserted this patent.
- **Samsung's own products excluded** per task instructions, even though Bothie-style Samsung Galaxy features (Dual Capture, Director's View) would otherwise be the most on-point technological references.

- **Geographic enforceability.** Without a confirmed US family member, US-only acts (e.g., Snap Inc.'s US operations) are not reachable through this patent. The Indian-market Snapchat user base is meaningful and directly reachable; that is where the enforcement focus, if any, must lie.